

Exact calibration repair after changed prefixes

Response moments, certified execution, and canonical deletion state

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Main advance. Repair no longer requires an unchanged first layer. We generate new quantized codes, bound the resulting retained activation drift, and certify those codes against the complete sequential retained-data quantizer. Ambiguous stages trigger selected retained-group replay. Complete replay is an exact fallback.

The central theorem concerns a *new certified prefix*. The new response-moment extension moves the surrogate toward that prefix using deletable polynomial statistics, with a conditional second-order remainder (Sections 17–21). A sharper covariance bound ignores harmless global scaling. A corpus-independent reference index gives canonical repeated-deletion state while preserving the original sequential model target. A weighted work scheduler bounds worst-case regression without predicting fresh runtime.

Previous unresolved point	Resolution in this revision
Early codes change	Certify transported new-prefix decisions; replay where the enclosure is insufficient.
Global minimum margin	Local cell checks; target-factor repair injects actual changed codes.
Trace scores too loose	Spectral intervals and covariance shape remove trace slack and harmless scaling.
Repeated/canonical state	Additive statistics of independent references; discard historical calibrated anchors.
Unknown fallback cost	Weighted dovetail gives a deterministic charged-work bound.
Universal fast repair	A specified feature-query lower bound rules it out without structural assumptions.
Practical speed/quality	Remain empirical questions. Software correctness checks are separate from experiments.

Scope of closure. Correctness, finite fallback, state semantics, and conditional cost guarantees can be closed mathematically. No theorem establishes favorable certificate coverage for arbitrary language models or reliable measured latency. Calling those empirical questions closed would be unjustified.

Evidence provenance. Earlier source/log/PDF files were removed by workspace maintenance before this turn. Historical findings in the repository are explicitly reconstructed from conversation context. This document contains new derivations and review, not a re-execution or recovery of those experiments.

1. Fix the target and the persistent state

Let R be retained records and $\ell = 1, \dots, L$ quantization stages in a deterministic acyclic graph. Feature matrix $X_{\ell j}$ of record j is evaluated using already quantized ancestors. The primary model target is

$$Q_{\text{seq}}(W, R), \quad H_{\ell}(R) = \lambda_{\ell} I + \frac{1}{M_0} \sum_{j \in R} X_{\ell j} X_{\ell j}^{\top}, \quad \lambda_{\ell} > 0.$$

Original weights W , grids, order, tie rule, normalization M_0 , feature program and graph are fixed. Record-local evaluation excludes unaccounted corpus-dependent packing/batching. Relative damping and other branches are treated in Sections 3 and 12.

The exact statistic/rounding oracle V treats finite feature-program outputs as exact dyadic values, forms exact moments, and performs deterministic rational metric rounding. Local fallback uses rational elimination and exact grid comparisons. The previously measured floating reduction/Cholesky executable E is a different target. Equality to V does not automatically establish bitwise equality to E .

Neural evaluator obligation. Transport must enclose the actual declared feature evaluator. Operator inequalities in Section 5 are directly valid for ideal operators. For a pinned floating program, add sound numerical-error bounds for both reference and candidate executions, or enclose that program directly. Unverified error tolerances are not certificates.

Fix reference networks $P_1, \dots, P_B$ *independently of the deletable corpus*. The original W , deterministic weight-only quantization of W , or independently specified proxies are allowed. Freezing a model fitted to the original calibration corpus does not make it independent.

For each reference retain intrinsic per-record summaries and exact additive moments. Define the complete promised state as

$$\mathcal{A}(R) = (Q_{\text{seq}}(W, R), I_{\text{ref}}(R), \text{IDs}(R)). \quad (1)$$

The fresh constructor discards true sequential Grams and calibration traces after building the model. Index I_{ref} is a specified canonical function of retained records and fixed references. This is a *new auxiliary-state interface with the same model target*. It is not equality to a historical interface persisting all true sequential activations.

The service may use an old model as a transient proposal, but every old anchor, ticket, candidate trace and deleted entry must be removed or regenerated according to (1) before committing. Canonical mathematical sums need canonical serialization if byte equality is promised. The claim does not remove knowledge originally learned in the fixed base weights W .

Retained records or sufficient tokenized inputs remain available through a declared retained-only source accessible to both baseline and repair, with reads charged. IDs and reference moments alone do not guarantee the ability to replay an uncertain nonlinear stage.

2. A sharp bound depending on covariance shape

For an SPD surrogate H , factor

$$H = \mathcal{L}^\top D^{-1} \mathcal{L}, \quad \mathcal{L} = I - B, \quad D = \text{diag}(g), \quad g_i > 0.$$

For row w and *any forced preceding code prefix* $q_{<i}$, write

$$v_i = w_i - B_{i,<i}(w - q)_{<i}, \quad E_i = \sum_{h<i} \frac{(v_h - q_h)^2}{g_h}, \quad K_i = \sqrt{g_i E_i}.$$

Use the same forced prefix in both quadratics; it need not be the deployed prefix.

Theorem 1 (Scale-aware conditional stability). *If $0 < a \leq b$ and $aH \preceq H_* \preceq bH$, then*

$$\boxed{|v_i^* - v_i| \leq \chi(a, b) K_i, \quad \chi(a, b) = \frac{b - a}{2\sqrt{ab}}.} \quad (2)$$

The coefficient is sharp for the abstract quadratic perturbation class. Its matrix geometry is classical; the contribution here is the forced-prefix quantization certificate, not a new universal matrix inequality.

Proof. Positive scaling does not change a quadratic's constrained minimizer. Rescale H_* by $2/(a+b)$ and whiten H . The relative perturbation A has norm at most $\eta = (b - a)/(b + a) < 1$. The old minimum-energy residual u is perpendicular to the remaining adjustable subspace V , with $\|u\|^2 = E_i$. Put $D_0 = A_{VV}$ and $C = A_{V,V^\perp}$. The new residual is $u + y$, where

$$y = -(I + D_0)^{-1} C u, \quad C C^\top \preceq \eta^2 I - D_0^2.$$

The squared operator norm is bounded by

$$\max_{z \in [-\eta, \eta]} \frac{\eta^2 - z^2}{(1 + z)^2} = \frac{\eta^2}{1 - \eta^2},$$

attained at $z = -\eta^2$. The coordinate functional has squared dual norm g_i . Cauchy–Schwarz proves (2). A two-dimensional block with eigenvalues $\pm\eta$ attains this operator bound. $\square$

Prior geometry. This coefficient follows from Kantorovich/Wielandt angle bounds; see source [7] and the explicit reduction in `docs/NOVELTY_AUDIT.md`. We include a direct proof to fix the quantization conventions.

This recovers $\rho/(2\sqrt{1-\rho})$ for PSD deletion with $(a, b) = (1 - \rho, 1)$, and gives the sharper signed bound $\eta/\sqrt{1-\eta^2}$. If $a = b$, every code is unchanged even under a large proportional covariance change.

An interval $[v_i - \chi K_i, v_i + \chi K_i]$ inside one rounding cell certifies that cell. Check coordinates inductively. Zero radius handles ties using the exact tie rule; positive-radius boundary contact abstains. Saturated cells have one finite boundary. Squared rational comparisons avoid square roots. Under scaling $H \mapsto cH$, g_i scales by $1/c$ and E_i by c , so K_i is invariant.

3. Stronger deletion scores and relative damping

At an input-stable stage let $H' = H - \sum_{j \in F} \Delta_j$, with $\Delta_j \succeq 0$. Replace trace leverage with certified spectral intervals

$$l_j I \preceq H^{-1/2} \Delta_j H^{-1/2} \preceq u_j I.$$

Sum requested endpoints: $l_F = \sum_F l_j$, $u_F = \sum_F u_j$. If $u_F < 1$, (2) applies with

$$a = 1 - u_F, \quad b = 1 - l_F, \quad \chi = \frac{u_F - l_F}{2\sqrt{(1 - u_F)(1 - l_F)}}. \quad (3)$$

Use outward rational/dyadic endpoints. A scalar maximum-score gate remains available with $l_j = 0$ and upper operator bounds u_j . Keeping both endpoints gives $O(kL)$ online work and $O(NL)$ interval storage. Preparation, authentication and precision are charged.

Why this matters. When $\Delta_j = H/N$, trace leverage is d/N , whereas $l_j = u_j = 1/N$. Formula (3) has zero shape error for every proper proportional deletion. The old trace gate loses a factor d and misses exact scale invariance. This is an algebraic regime, not an empirical synthetic dataset. Low-rank records often have $l_j = 0$, so this improvement is not universal.

Fixed positive ridge does not make $\Delta_j = H/N$ automatic. The example is compatible with the following additive relative-damping contract, or another contribution family satisfying that equality.

For trace-proportional damping use unnormalized statistics

$$S = \sum_j X_j X_j^\top, \quad H = S + \frac{\zeta}{d} \text{tr}(S) I.$$

Deleting a record removes the PSD contribution

$$\Delta_j = X_j X_j^\top + \frac{\zeta}{d} \|X_j\|_F^2 I. \quad (4)$$

Refitting this damping rule is exactly additive. Changing total to mean normalization multiplies the whole damped metric by a positive scalar, leaving exact rounding unchanged. If a fixed ridge floor is also present, retain it explicitly; scale cancellation need not hold for a differently normalized contract.

After changed ancestors, transport the retained damping term too. For $\zeta \geq 0$,

$$\Delta H = \Delta S + (\zeta/d) \text{tr}(\Delta S) I, \quad \|\Delta H\|_2 \leq (1 + \zeta) \|\Delta S\|_2.$$

The group Frobenius bounds δ_G in Section 6 bound both spectral error and absolute trace error, giving the sharper bound $(1 + \zeta/d) \sum_G \delta_G$ when applicable. Simple deleted-Gram subtraction is insufficient after retained inputs change.

Dead-feature, zero-data, order and grid branches must be fixed or certified. Pure relative damping is not SPD on zero data: specify a floor or deterministic fallback. Floating programs need not inherit exact scale invariance.

These scores refer to one specified anchor metric. Do not reuse them after earlier deletions as though they use the current inverse. The canonical reference design removes such corpus-conditioned scores from committed state, or recomputes them if the declared state interface includes them.

4. Exact repair after the first stage changes

Assume repaired ancestors of stage ℓ are certified equal to $Q_{\text{seq}}(W, R)$. Let X_* concatenate that target's retained features. Let Z be reference or partially replayed features, and put

$$\bar{H} = \lambda I + ZZ^\top / M_0, \quad E = X_* - Z, \quad a_0 = \left\| \bar{H}^{-1/2} Z / \sqrt{M_0} \right\|_2 \leq 1.$$

If $e \geq \left\| \bar{H}^{-1/2} E / \sqrt{M_0} \right\|_2$, then

$$\left\| \bar{H}^{-1/2} (H_* - \bar{H}) \bar{H}^{-1/2} \right\|_2 \leq 2a_0 e + e^2 =: \eta. \quad (5)$$

Expand $ZE^\top + EZ^\top + EE^\top$ and apply submultiplicativity. One conservative choice is

$$e = \|E\|_F / \sqrt{M_0 \lambda},$$

or a certified upper bound on this value. Tighter eigenvalue enclosures may supply (a, b) directly. For $\eta < 1$, use $(a, b) = (1 - \eta, 1 + \eta)$.

Theorem 2 (Transported sequential repair). *Process the quantization DAG topologically. At each stage form a surrogate metric and a sound relative spectral enclosure for the true retained metric under already certified new ancestors. Generate and certify every rounding decision by Theorem 1. If every stage accepts, the complete model equals $Q_{\text{seq}}(W, R)$, even if the first stage or every stage changes codes.*

Proof. The first stage's feature target is independent of quantized ancestors. At a general stage, induction identifies true retained features with those evaluated under the certified candidate prefix. The supplied enclosure therefore contains the true metric. Coordinate induction using (2) identifies every emitted code with the local target. This certified stage establishes the premise for descendants. Fixed nonquantized parameters complete equality. $\square$

No circular premise. We bound features using a prefix already proved correct. We do not assume the old prefix remains correct, or certify mutually dependent tentative stages simultaneously.

For a candidate trace let $\tau = \min_{K_i > 0} m_i / K_i$, where m_i is its inside-cell boundary slack. A sufficient check is

$$\eta < \frac{\tau}{\sqrt{1 + \tau^2}}. \quad (6)$$

Local cell checks are preferable to abandoning all reuse at one small margin. The candidate may differ from the old model. Failure proves neither incorrectness nor unavoidable replay: another anchor, tighter enclosure, or selected group replay may resolve it.

5. Transport for transformer primitives

Maintain group reference norms and certified candidate-minus-reference errors. Aggregate Frobenius bounds avoid visiting each retained record. Matrix orientations below must be adapted consistently to the implementation.

Affine and residual nodes. For $Y = WX$, $Y' = W'X'$,

$$\|Y' - Y\|_F \leq \|W'\|_2 \|X' - X\|_F + \|W' - W\|_2 \|X\|_F.$$

Bias changes add their broadcast Frobenius norm; residual sums add parent bounds. A K -Lipschitz elementwise map multiplies its input bound by K , on a certified domain containing both executions.

Layer normalization. For width d , epsilon $\epsilon > 0$, fixed gain γ , reference centered RMS at least s , and per-token perturbation at most b , a segment-valid Lipschitz bound is

$$L_{\text{LN}} \leq \frac{\|\gamma\|_\infty}{\sqrt{\max\{s - b/\sqrt{d}, 0\}^2 + \epsilon}}. \quad (7)$$

Centering is an orthogonal projector; differentiate the normalized centered vector and bound the denominator along the segment. A group Frobenius error bounds every token conservatively. Gain/bias changes add terms. Deleted minima/maxima require retained multiset/tree maintenance; stale extrema are not canonical state.

Attention. For one head $S = QK^\top/\sqrt{d_h}$ with fixed mask,

$$\|S' - S\|_F \leq \frac{\|Q'\|_2 \|K' - K\|_F + \|Q' - Q\|_F \|K\|_2}{\sqrt{d_h}}.$$

The row-softmax Jacobian $\text{diag}(p) - pp^\top$ has operator norm at most $1/2$, since its absolute row sums are $2p_i(1 - p_i) \leq 1/2$. Integration gives

$$\|A' - A\|_F \leq \frac{1}{2} \|S' - S\|_F, \quad \|A'V' - AV\|_F \leq \|A'\|_2 \|V' - V\|_F + \|A' - A\|_F \|V\|_2. \quad (8)$$

For t tokens, $\|A'\|_2 \leq \|A'\|_F \leq \sqrt{t}$. Tighter bounds are allowed. Include projections, head combination, rotary maps and all residual edges. Mask supports must match; fully masked rows need the declared fallback.

Floating execution. If candidate/reference finite node evaluations deviate from their ideal operations by at most ν', ν on the certified domains, add $\nu' + \nu$ to the ideal difference bound. Prove these for the actual kernels and reductions. If no sound envelope is available, replay the required records or use a verified primitive. Mathematical Lipschitz bounds alone do not certify arbitrary GPU kernels.

6. Adaptive replay and finite fallback

Use a corpus-independent partition or fixed ID tree. For each reference, stage and stored group G , retain reference Gram S_G^0 and intrinsic energy/domain summaries. Under a fixed certified new prefix, bound

$$\|X_G^* - Z_G\|_F \leq e_G, \quad \delta_G = \frac{2\|Z_G\|_F e_G + e_G^2}{M_0}.$$

An unreplayed group has $\|H_G^* - \bar{H}_G\|_2 \leq \delta_G$. A replayed group uses its exact target contribution and zero uncertainty. Thus

$$\|H_* - \bar{H}\|_2 \leq \sum_{G \text{ unreplayed}} \delta_G, \quad \eta \leq \frac{\sum_G \delta_G}{\lambda}. \quad (9)$$

This conservative absolute uncertainty is monotone under replay. Sharper whitened or anisotropic enclosures may supplement it.

Step Stage-local algorithm

- 1 Freeze certified new ancestors. Select an available reference. Initialize the retained reference Gram.
 - 2 Propagate group errors and build a sound metric enclosure. Factor the surrogate; generate and certify candidate decisions.
 - 3 Accept if all decisions and auxiliary branches are certified. Otherwise tighten precision or select an unresolved group for replay.
 - 4 Evaluate selected records under the fixed certified prefix. Replace their reference contributions with exact target contributions and remove their uncertainty. Retry the current candidate.
 - 5 After all groups are replayed, execute the exact rational local quantizer and advance.
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Theorem 3 (Soundness and termination). *With finite groups, a progress rule that eventually replays a new unresolved group, and a terminating exact local oracle, the algorithm returns $Q_{\text{seq}}(W, R)$ and terminates even at ties. Certified ancestors stay fixed; changing a current candidate invalidates tentative descendants.*

Proof. Equation (9) encloses unknown contributions at each retry; acceptance follows from Theorem 2. After finitely many strict replay steps every uncertainty is zero and the exact local oracle terminates. Induct over the finite DAG. Limit precision-only retries so they cannot prevent replay progress. $\square$

Selecting groups by uncertainty removed per predicted work is a heuristic, not an optimality theorem. Ranking, heap reads, refactorization, repeated certification and duplicated forwards all count. Cache replayed intermediate activations only transiently, keyed by the exact certified prefix. Reference replay is also required when a group’s reference Gram was not stored.

7. Canonical state under repeated deletion

For reference b , stage ℓ and stored group G , the persistent index keeps

$$S_{\ell G}^b(R) = \sum_{j \in R \cap G} X_{\ell j}^b X_{\ell j}^{b\top}.$$

Exact subtraction of removed reference contributions updates the moments. Removed content must be available before erasure, or its intrinsic contribution must have been stored. An ID alone cannot reconstruct an absent contribution.

Theorem 4 (Canonical service with the original model target). *Let the fresh constructor output exactly (1), using fixed reference maps and canonical index construction/serialization. Update the index on deletion, compute the target model by transported repair or exact fallback, and discard noncanonical traces. The committed result is exactly $\mathcal{A}(R \setminus F)$. Any deletion sequence gives fresh state for its final retained set.*

Proof. Reference moments are sums of intrinsic record functions, so subtraction gives their defining retained values. Recompute extrema and map/tree metadata canonically. Theorems 2–3 or fallback supply the target model. Erasure leaves precisely (1). Induction over requests proves order independence. $\square$

Use a fixed ID-key universe or a deterministically rebuilt retained-ID structure. Empty nodes encoding deleted IDs violate byte-canonical state if fresh construction omits them. Fixed hash-prefix buckets with canonical retained maps are one option. Balance/index rules must be specified; set equality alone is not serialized-state equality.

With B references and G stored groups, Gram storage is

$$O\left(BG \sum_{\ell} d_{\ell}^2\right)$$

arithmetic values, plus intrinsic descriptors, IDs and exact precision. Refinement without a stored group Gram must recover that contribution, possibly by reference replay. A full tree can cost $O(BN \sum d_{\ell}^2)$ and is not a cheap default. Select coarse groups under an explicit memory budget.

Bank monotonicity. Adding an independent reference weakly enlarges the union of certificate routes: an optimal selector can still choose the old route. This does not prove latency monotonicity for an implemented selector. Extra references cost setup, storage, deleted-side updates and selection.

Coverage boundary. Reference transport measures full candidate-versus-reference drift. It need not shrink as $|F|/|R| \rightarrow 0$; a gate may abstain even for an empty deletion. The old small-deletion acceptance theorem cannot be transferred without a new coverage hypothesis. If the interface instead persists true sequential Grams, their refresh must be charged separately. Historical inverse-leverage tables are not intrinsic reference descriptors.

8. Local repair once a target factor exists

When exact target factor B' is available, avoid restarting each changed row's whole suffix. With old trace (q_i, v_i) define

$$b_i = -(B' - B)_{i, < i}(w - q)_{< i}, \quad |b_i| \leq \epsilon_i, \quad d_i = \sum_{h < i} B'_{ih}(q'_h - q_h).$$

The true input is

$$v'_i = v_i + b_i + d_i. \quad (10)$$

A spectral perturbation bound under the old forced prefix supplies ϵ_i . Maintain d_i explicitly as earlier code changes are discovered.

At coordinate i , test whether $[v_i + d_i - \epsilon_i, v_i + d_i + \epsilon_i]$ lies in one cell. If so, emit that cell, which may differ from the old code. Otherwise compute the exact target input from B' and the repaired prefix. If $q'_i \neq q_i$, inject

$$d_j \leftarrow d_j + B'_{ji}(q'_i - q_i) \quad (j > i).$$

Theorem 5 (Sparse change injection). *Given exact target factors, the old trace and sound ϵ_i , this procedure returns exact target codes. For p rows of width d , s changed decisions and r ambiguous decisions, arithmetic work is*

$$O(pd + (s + r)d), \quad (11)$$

excluding factor acquisition, trace/envelope construction and output.

Proof. Splitting $w - q'$ into $w - q - (q' - q)$ gives (10). Correct previous codes make either containment or exact recomputation sound. Injection maintains (10) for later coordinates. Induct over coordinates and rows. There are pd checks, at most d work per ambiguous exact dot product and d updates per changed code. $\square$

This avoids a global minimum-margin requirement for the local repair algorithm. Near-boundary decisions incur exact work instead of invalidating all reuse. Worst-case $s + r = \Theta(pd)$ recovers a dense sweep. Neither small perturbations nor few changed input records guarantee few changed codes.

Two distinct regimes. In transported repair the true target factor may be unknown. An ambiguous coordinate cannot then be cheaply recomputed from it. Tighten the metric enclosure, replay records, or fall back. Formula (11) applies only after the target factor exists. Old calibrated traces are transient under (1); keeping them permanently changes the state contract. The exact core implements the shape verifier; `src/sparse_repair.py` implements this sparse recurrence, with factor/trace/envelope premises explicitly labeled.

9. Full-model cost with changed codes

Compare identical output/state interfaces, numerical oracles and storage allowances. Give the fresh/indexed baseline the reference statistics and caches given to repair; it may reuse them wherever valid. A deliberately scanning fresh constructor is not a universal lower bound.

Let baseline disjoint costs be retained feature/statistic work $F > 0$, factor/rounding work G , and common service/output work J . Suppose repair replays at most fraction s of F , uses one quantization schedule bounded by G , and adds work U :

$$C_{\text{base}} = F + G + J, \quad C_{\text{repair}} \leq sF + G + J + U. \quad (12)$$

U includes deleted-side reference inference, moment updates, transport, extra candidate sweeps and factorizations, numerical validation, selection, duplicate forwards, extra reads and canonical commit/purge. If verification costs more than the shared G schedule, charge its excess in U .

Theorem 6 (Whole-model charged-work gain). *If $s + u < 1$, $U/F \leq u$ and $(G + J)/F \leq \gamma$, then*

$$\boxed{\frac{C_{\text{base}}}{C_{\text{repair}}} \geq \frac{1 + \gamma}{s + u + \gamma} > 1.} \quad (13)$$

There is no restriction on the number of changed model codes.

Proof. For $g = (G + J)/F$, (12) gives ratio at least $(1 + g)/(s + u + g)$. Its derivative in g is $(s + u - 1)/(s + u + g)^2 < 0$. Substitute $g \leq \gamma$. Exactness follows separately from Theorems 2–4. $\square$

This is an operational sufficient condition with a complete ledger. A class satisfying it on every request gets deterministic per-request savings. A probability claim also needs a request law and a justified probability of satisfying it. No such law is assumed for targeted withdrawal.

The unchanged-prefix result is a special case. Here s counts actual retained recomputation, not changed layers, so changed first-stage weights are permitted. If all retained records are replayed and $U > 0$, the theorem correctly cannot show a saving.

Preprocessing P_0 is additional lifetime work. Net lifetime savings require $\sum_r (C_{\text{base},r} - C_{\text{repair},r}) > P_0$. Materializing a P -word model costs $\Omega(P)$ writes. Unit-cost rational arithmetic is not bit-runtime: precision, memory bandwidth and serialization must be charged before claiming latency. No favorable numerical constants in (13) have been measured for the new algorithm.

10. Fallback without predicting fresh runtime

Replace a cap requiring unknown fresh runtime with weighted dovetail. Let O be a terminating exact baseline and R a certified repair service. Both implement the same complete state contract. Run isolated resumable branches with work shares

$$w_O = \frac{1}{1 + \alpha}, \quad w_R = \frac{\alpha}{1 + \alpha}, \quad \alpha > 0.$$

Stop at the first certified complete artifact/state and commit once. Branch-local verification and normal cleanup belong in branch costs; shared commit work is J . Additional cancellation of the losing branch costs at most D_{cancel} .

Theorem 7 (Forecast-free bounded regression). *In a divisible charged-work model, if baseline and repair require branch work B and R , total work obeys*

$$C \leq J + \min\{(1 + \alpha)B, (1 + 1/\alpha)R\} + D_{\text{cancel}}. \quad (14)$$

Thus $C \leq (1 + \alpha)(B + J) + D_{\text{cancel}}$ even if repair never succeeds. The pure multiplicative bound requires cancellation to have already been paid inside the declared charged-work model.

Proof. After total branch work T , each unfinished branch has its share. Baseline completes by B/w_O , repair by R/w_R if successful. Stop at first completion; add common commit and cancellation work. $\square$

For indivisible operations, specify a weighted-fair packet scheduler. If each branch's allocated work differs from its fluid share by at most q , then

$$C \leq J + \min\{(B + q)/w_O, (R + q)/w_R\} + D_{\text{cancel}}. \quad (15)$$

An uninterruptible factorization can make q large. Suspension, bookkeeping and memory pressure are costs; this is not an automatic GPU latency bound.

Cancellation accounting. Bounded arena disposal is still additional work, not automatically common baseline overhead. Reserve cleanup credits as part of each branch's comparable amortized work, or keep the explicit D_{cancel} . If $D_{\text{cancel}} \leq \delta(B + J)$, the fluid service is $(1 + \alpha + \delta)$ -competitive. Both branches must finish the same canonical state.

For multiple routes with shares $w_i > 0$, $\sum_i w_i = 1$, the same argument bounds branch work by $\min_i C_i/w_i$ before common/cancellation costs. Keep a positive share for an exact terminating oracle. Charge duplicated index construction and output.

Returned outputs are exact. With zero extra cancellation/packet cost, strict gain requires $R < \alpha B/(1 + \alpha)$; otherwise include that overhead too. Scheduling cannot create a gain when every valid route does fresh work. Fix shares before evaluation or account for adaptive selection.

11. Limits on unconditional retained-free repair

Rank can grow through attention. Start with $n \times n$ zero logits. For distinct positive t_i , take $u_i = \log t_i$, $v_j = j - 1$. New logits $uv^\top$ have rank one, but their entrywise exponential is the nonsingular Vandermonde matrix t_i^{j-1} . Row normalization preserves rank, so new attention has rank n . Old attention has rank one; the difference has rank exactly $n - 1$ since row sums vanish. Low-rank logit changes do not guarantee low-rank downstream corrections.

Fixed polynomial moments do not suffice. For every m , equal-size finite multisets have matching moments through degree m but different sums of $f(x) = \text{GELU}(x)^2$. Choose a nonzero $(m+1)$ st finite difference of this analytic nonpolynomial f . Split its alternating binomial coefficients into positive/negative multiplicities. Finite differences annihilate degree- m polynomials but not this f . This rules out universal fixed-polynomial-moment closure, not every summary.

Theorem 8 (Feature-query obstruction). *Suppose preprocessing supplies old features and metadata independent of new feature-oracle outputs. A changed prefix is accessible only through a per-retained-record oracle, with no structural certificates coupling new outputs. An always-exact repair algorithm needs n queries in the worst case for n retained records, even for a two-coordinate suffix quantizer.*

Proof. Old features are zero. New features are either all zero or zero except an unknown record equal to $(1, 1)$. Put

$$H = \lambda I + M_0^{-1} \sum_j z_j z_j^\top, \quad r = (M_0 \lambda + 1)^{-1}, \quad w_1 = c \in (0, 1/2), \quad w_2 = 1/2 - cr/2.$$

On the integer grid the first code is zero. The second input $w_2 + cH_{12}/H_{22}$ rounds to zero in the all-zero case and one in the single-nonzero case, with strict margins. If record i is unqueried on the all-zero transcript, those cases are indistinguishable but require different artifacts. Query every record. Zero-error randomized algorithms obey this almost surely. With per-instance error ϵ , coupled random bits give expected queries at least $n(1 - 2\epsilon)$. $\square$

This is a lower bound for the stated feature-access model, not the computational complexity of an explicit transformer. Transport escapes by adding structural relations between new and reference features. These are mathematical constructions, not synthetic empirical datasets. Dense code changes alone do not imply a lower bound: linear feature maps admit exact low-order moment closure.

12. Numerical decisions and additional branches

For finite neural outputs and rational/dyadic W , grids and ridge, H is rational. Reverse unit-LLDL uses rational arithmetic; conditional inputs and grid comparisons are rational. Exact fallback therefore resolves midpoint ties finitely; endless interval refinement at a tie is unnecessary.

For $H = \mathcal{L}^\top \text{diag}(t)\mathcal{L}$, eliminate pivots from the last coordinate backward. Each positive t_i yields $\mathcal{L}_{i,<i}$ and a Schur update. Set $g_i = 1/t_i$, $B_{i,<i} = -\mathcal{L}_{i,<i}$. This supplies our geometry without asserting a rounded Cholesky factor is exact.

An accelerated floating factor defines a surrogate. Enclose its residual against the intended exact metric, compose with transport, and apply (2). Exact accumulator bins or normalized rational serialization make sums canonical. Ordinary floating subtraction is not order-independent exact state.

Target feature	Correct handling
Activation-order sorting	Certify the selected comparison/permutation, or acquire exact statistics and use the target tie rule.
Fitted scales/grids	Certify complete fitting decisions or recompute them. Fixed-grid proofs do not cover changed grids.
Dead-column rules	Certify zero/nonzero tests and weight resets; positive ridge does not remove a separate dead-column branch.
Relative damping	Update by (4), with a declared zero-data/floor policy.
Batching/packing	Fix record-local semantics or include cross-record operations in the dependency graph.
Autotuned kernels	Pin and enclose numerical behavior, or state that the exact-statistic oracle is a distinct target.

Reference core. The accompanying Fraction-only implementation supplies reverse LDL, sequential rounding, conditional spectral-shape certification and deletion-interval summation. It rejects floating inputs and marks spectral enclosures as caller-proved premises. This local module does not itself certify neural transport. The revision-3 service and decoder compose it with explicit providers and exact replay. Independent constrained-quadratic software tests, static compilation and manual review were performed; research experiments remain paused.

The accompanying `docs/NUMERICAL_CONTRACT.md` specifies analytic affine rounding bounds, interval primitive requirements, denominator checks and mandatory unknown/replay behavior. These are component guarantees, not machine-checked proofs of the Python implementation. Independent oracle checks must precede performance claims when execution resumes.

13. A separate margin-free alternative target

If original-target transport coverage is poor, use a separately named fixed teacher/proxy calibration quantizer. This is a different reference algorithm, not an undisclosed change to Q_{seq} .

For corpus-independent maps (X_j, Y_j) set

$$H = \lambda I + \frac{1}{M_0} \sum_j X_j X_j^\top, \quad A = \lambda W + \frac{1}{M_0} \sum_j Y_j X_j^\top, \quad U = AH^{-1}.$$

Its reconstruction objective completes the square:

$$\frac{1}{M_0} \sum_j \|QX_j - Y_j\|_F^2 + \lambda \|Q - W\|_F^2 = \text{tr}((Q - U)H(Q - U)^\top) + \text{const.}$$

Apply deterministic metric rounding to (U, H) independently at each module. Greedy rounding is not claimed to solve the discrete global minimum. With teacher inputs and $Y_j = WX_j$, $U = W$.

Theorem 9 (Deletion-native exact repair). *Subtract deleted second/cross moments, recompute U , and rerun declared local quantizers. The entire model and canonical moment state equal fresh construction on retained data after every deletion sequence, without a margin condition. No retained neural forward passes are needed, even if every emitted tensor changes.*

Proof. Features are intrinsic to records and independent of emitted quantized ancestors. Exact subtraction gives retained moments; positive ridge and deterministic rounding give the retained model. Induct over requests. $\square$

An exact center-update identity is

$$U' = U + (UX_F - Y_F)X_F^\top H'^{-1}/M_0. \quad (16)$$

Multiply by $H' = H - X_F X_F^\top / M_0$ to verify it. Low-rank solves/downdates are optional; dense refactoring remains fallback.

Let f be per-record fixed-feature/moment work, n retained and k deleted. Fresh costs $nf + G + J$, repair at most $kf + G + J + D$. If $k/n \leq \eta$, $(G + J)/(nf) \leq \gamma$, $D/(nf) \leq \delta$ and $\eta + \delta < 1$, the ratio is at least

$$\frac{1 + \gamma}{\eta + \gamma + \delta} > 1. \quad (17)$$

This is margin-free but target-specific. Give fresh the same cache and index; charge precision and storage. Teacher-input and asymmetric reconstruction already exist, so the contribution cannot be anchoring alone.

14. Quality accounting for the alternative

For mathematical layers $f_\ell(x) = \sigma_\ell(W_\ell x)$ assume σ_ℓ is L_ℓ -Lipschitz on a domain containing teacher and deployed activations. On a specified set use RMS Frobenius norm $\|Z\|_S = \|Z\|_F / \sqrt{M}$. Let $T_{\ell-1}$ be teacher input, $e_{\ell-1}$ deployed-teacher RMS error, and $E_\ell = Q_\ell - W_\ell$. Define

$$R_\ell = \|E_\ell T_{\ell-1}\|_S^2 + \lambda_\ell \|E_\ell\|_F^2.$$

For $\|D\|_S \leq d$, triangle inequality and Cauchy–Schwarz give

$$\|E_\ell(T_{\ell-1} + D)\|_S \leq \sqrt{(1 + d^2/\lambda_\ell)R_\ell}.$$

Hence

$$e_\ell \leq L_\ell \left[\|W_\ell\|_2 e_{\ell-1} + \sqrt{(1 + e_{\ell-1}^2/\lambda_\ell)R_\ell} \right], \quad e_0 = 0. \quad (18)$$

Another valid bound is $L_\ell[\|Q_\ell\|_2 e_{\ell-1} + \|E_\ell T_{\ell-1}\|_S]$; use the smaller valid bound. Numerical teacher/deployment errors add explicit terms. Residual/attention architectures need graph recurrences; arbitrary transformer operations are not globally one-Lipschitz.

For final logits, mean token cross-entropy differs by at most $\sqrt{2}e_{\text{final}}$ on this set: cross-entropy’s logit gradient has norm at most $\sqrt{2}$. This is a deterministic set-level bound, not a population accuracy or perplexity guarantee. Large products of Lipschitz constants may make it loose.

Block reset alternative. Fix teacher calibration anchors at chosen block entrances; retain sequential calibration within each block. Deployment composes all quantized blocks normally. This confines propagation to block interiors but does not eliminate retained replay inside changed non-linear blocks. One full-model block recovers sequential calibration; one-stage blocks recover the teacher-input special case, not arbitrary asymmetric proxies.

For teacher-entrance discrepancy ϵ_b and deployed block Lipschitz constant K_b , $e_b \leq K_b e_{b-1} + \epsilon_b$. Longer blocks do not universally improve this bound. A partition fitted on deletable data must itself be repaired; otherwise fix it independently.

The independent note gives an optional corpus-independent candidate-bank dynamic program optimizing a compositional quality bound under deployment budget. That is a narrower model-selection target, retained as a reserve. The primary recommendation remains original sequential repair with transported certificates.

15. Implementation contract and next phase

Primary algorithm: transported exact sequential repair. Its reference state preserves Q_{seq} and its stage verifier uses Theorem 1. The complete service requires these interfaces:

Component	Required contract
Reference index	Intrinsic summaries, exact subtraction, canonical IDs/tree/extrema, explicit memory budget.
Transport provider	Sound group errors for declared feature programs, including numerical envelopes.
Metric enclosure	Certified positive spectral endpoints relative to an exact or validated surrogate.
Decision engine	Fixed-grid rational oracle; accelerated interval route with exact tie fallback.
Replay engine	Certified ancestor identities, complete dependencies, deterministic record-local evaluation.
Scheduler	Resumable charged-work steps, packet-lag bound, always-terminating exact branch.
Commit	Complete artifact/state verification, canonical serialization and transient/deleted-data erasure.

The reference implementation includes the local core, sparse injection, response moments, canonical stage service, deterministic decoder, and cooperative scheduler. Revision 4 adds compact aggregate state, automatic certified transformer responses, and local GPT-2 checkpoint loading. Sections 24–26 define their scope. The certified decoder defines a separate finite program, V_{cert} . It does not certify arbitrary library kernels. This is new code, not recovery of the missing implementation.

The shape lemma, new-prefix induction, adaptive fallback, state constructor, sparse local repair, full-model cost condition and weighted fallback were derived and independently reviewed. Notes preserve the derivations and challenges. Python code remains a reference implementation, not a formal proof.

When experiments are authorized again, first establish whether validated transport saves enough retained replay to exceed verification/index costs against a baseline with identical state access. Keep changed-first-layer cases, failed certificates and fallbacks in the workload. The decisive result is saved full-model work, not a rounding-only kernel gain.

No research benchmark, synthetic dataset study, model download or external compute job ran during this revision. Software correctness tests use small algebraic fixtures to check implementation behavior; they do not establish model quality, useful certificate coverage or speed.

Continuity. The repository at <https://github.com/priyankjairaj100/second> contains restart context, historical provenance, source, specifications and independent notes. Read `RESUME.md` and `docs/STATUS.md` before continuing. Missing historical files are inventoried, not recreated as if original.

16. Paper claims and primary sources

The defensible contribution is certified calibration-data removal through a sequential quantization pipeline *with changed ancestors*, canonical reference state, and an explicit cost boundary. A paper needs a measured saved-replay phenomenon. A conditional theorem or a different teacher-calibrated target does not establish that phenomenon.

Claim	Disposition
Exact changed-prefix output	Proved on certificate acceptance; exact fallback always returns the declared target.
Order-independent state	Proved for the reference-index constructor (1).
Bounded regression	Proved in the work model, with packet and cancellation costs.
Every small deletion is fast	Not implied; reference drift need not shrink with deletion size.
Universal retained-free repair	Ruled out in the specified black-box model, not for all explicit transformer algorithms.
Practical latency gain	Not established; requires execution and complete cost accounting.
Anchoring itself is novel	Not claimed. Existing sources cover this ingredient.

Primary sources checked on 4 October 2026.

- [1] Frantar et al. *GPTQ: Accurate Post-Training Quantization for Generative Pre-trained Transformers*. ICLR 2023. <https://arxiv.org/abs/2210.17323>; <https://github.com/IST-DASLab/gptq>. Second-order quantization is baseline work. The within-block true-sequential option does not remove inter-block propagation.
- [2] Chen et al. *The Geometry of LLM Quantization: GPTQ as Babai's Nearest Plane Algorithm*. ICLR 2026. <https://github.com/IST-DASLab/GPTQ-Babai>. The geometric interpretation is prior work; the present bounds address deletion and transport.
- [3] Li et al. *GPTAQ: Efficient Finetuning-Free Quantization for Asymmetric Calibration*, earlier GPTQv2. <https://arxiv.org/abs/2504.02692>. Asymmetric reconstruction and accumulated calibration mismatch are established topics.
- [4] Li et al. *BRECQ: Pushing the Limit of Post-Training Quantization by Block Reconstruction*. ICLR 2021. <https://arxiv.org/abs/2102.05426>. Block reconstruction is prior work.
- [5] NVIDIA Model Optimizer, configuration source. <https://github.com/NVIDIA/Model-Optimizer/blob/main/modelopt/torch/quantization/config.py>. Setting `get_qdq_activations_from_prev_layer` to false supplies subsequent layers with full-precision activations. Independent-input calibration alone is not novelty.
- [6] Vaswani et al. *Attention Is All You Need*. <https://arxiv.org/abs/1706.03762>. Dot-product/softmax attention defines the analyzed operator.

This is a focused source check, not an exhaustive priority determination. The next phase tests the useful regime of the proved statements; experiments remain paused until the user instructs resumption.

17. Moving references through deletable response moments

Revision 3: the main algorithmic improvement. A fixed independent reference can have large error even for an empty deletion. We replace its fixed features by a parameter-responsive surrogate whose entire Gram can be queried from exact retained sums. This improves the local error order under explicit conditions; it does not assume that all quantization changes are small or low dimensional.

For a stage, let θ contain the dequantized parameters of its already certified ancestors. Before seeing the removable corpus, fix a reference θ_0 , directions $D_1, \dots, D_r$, a domain and all feature/derivative providers. Decompose

$$\theta = \theta_0 + Da + u, \quad Da = \sum_{t=1}^r a_t D_t.$$

The residual u is explicit. Directions fitted on the deletable corpus need their own repair theorem. Differentiation concerns the real feature map in its weight parameters, not the discontinuous quantizer producing those parameters.

Let record j have intrinsic rational feature jets $Z_{j0}, \dots, Z_{jr}$, and define

$$Z_j(a) = Z_{j0} + \sum_{t=1}^r a_t Z_{jt}, \quad c = (1, a_1, \dots, a_r).$$

For a retained group G , store symmetric coefficient matrices

$$\begin{aligned} T_{G,tt} &= M_0^{-1} \sum_{j \in G} Z_{jt} Z_{jt}^\top, \\ T_{G,tu} &= M_0^{-1} \sum_{j \in G} (Z_{jt} Z_{ju}^\top + Z_{ju} Z_{jt}^\top), \quad t < u. \end{aligned}$$

Theorem 10 (Exact response-moment closure). *With $K_r = \binom{r+2}{2}$ coefficient matrices, the surrogate Gram is exactly*

$$S_G(a) = \sum_{0 \leq t \leq u \leq r} c_t c_u T_{G,tu} = M_0^{-1} \sum_{j \in G} Z_j(a) Z_j(a)^\top. \quad (19)$$

It can be evaluated without retained record reads. Deleting records subtracts their intrinsic coefficients; the index equals fresh construction on the retained set after any deletion order.

Proof. Expand the finite product and combine transpose-related cross terms. Every coefficient is an exact sum of a fixed record-local function. Subtraction and induction yield the retained sum. Canonical rational serialization and retained-only metadata give equality of bytes. $\square$

Negative cross coefficients do not invalidate positive semidefiniteness: the complete expression is a sum of Grams. The implementation stores oriented $Z_s Z_t^\top$ for $s \leq t$ and symmetrizes during queries; it is constant-factor larger than the minimal symmetric representation above.

This index is a proposal mechanism for the original Q_{seq} . The approximation is never installed merely because it is close. It must pass the exact decision certificate, or the service replays retained groups.

18. A response remainder with no hidden error terms

Write $f_j(\theta)$ for the ideal real feature map and $V_j(\theta)$ for the pinned finite target. Assume uniform bounds on the declared chart and residual tube:

$$\begin{aligned} \|D^2[f_j(\theta_0 + D\cdot)](z)[v, w]\|_F &\leq H_j \|v\|_2 \|w\|_2, \\ \|Df_j(\theta_0 + Da + su)[u]\|_F &\leq L_j \|u\|_\Theta, \quad 0 \leq s \leq 1, \\ \|f_j(\theta_0) - Z_{j0}\|_F &\leq e_{j0}, \\ \|Df_j(\theta_0)[D_t] - Z_{jt}\|_F &\leq e_{jt}, \\ \|V_j(\theta) - f_j(\theta)\|_F &\leq \nu_j. \end{aligned}$$

The Hessian bound includes all mixed directional derivatives. The chosen norm on parameter residuals is part of the provider contract.

Theorem 11 (Sound second-order response enclosure). *Under these premises,*

$$\|V_j(\theta) - Z_j(a)\|_F \leq \nu_j + e_{j0} + \sum_t |a_t| e_{jt} + \frac{1}{2} H_j \|a\|_2^2 + L_j \|u\|_\Theta. \quad (20)$$

Proof. Insert the ideal endpoint, integrate its derivative along the residual segment, then apply Taylor's integral remainder along sa :

$$h_j(a) - h_j(0) - Dh_j(0)a = \int_0^1 (1-s) D^2 h_j(sa)[a, a] ds.$$

Add the constant and directional jet errors and the finite evaluator error. □

If Z_{j0} is the exact finite reference output, e_{j0} must still cover reference-versus-ideal error. Unverified automatic differentiation cannot supply certified jets. Floating programs are not globally smooth real maps; unsupported numerical providers return UNKNOWN and trigger replay.

Evaluate the error without scanning records. Define nonnegative descriptors

$$b_j = (\nu_j + e_{j0}, e_{j1}, \dots, e_{jr}, H_j, L_j), \quad v = (1, |a_1|, \dots, |a_r|, \|a\|^2/2, \|u\|_\Theta).$$

Store the additive matrix $C_G = M_0^{-1} \sum_{j \in G} b_j b_j^\top$. Then

$$\epsilon_G^2 := v^\top C_G v \geq M_0^{-1} \sum_{j \in G} \|V_j(\theta) - Z_j(a)\|_F^2. \quad (21)$$

The cross terms matter: summing only squared individual errors is unsound. Uniform intrinsic bounds enable exact deletion and canonical repeated state. Online record-dependent bounds would reintroduce retained reads. Rigorous upper bounds for irrational norms suffice because b_j, v are nonnegative.

19. Exact decisions and a larger local certified region

Set $S = \sum_G S_G(a)$, $\bar{H} = \lambda I + S$ and $\epsilon^2 = \sum_G \epsilon_G^2$. The existing transport theorem yields

$$\eta \geq \left\| \bar{H}^{-1/2} (H_* - \bar{H}) \bar{H}^{-1/2} \right\|_2, \quad \eta = 2a_0 e + e^2, \quad e = \epsilon / \sqrt{\lambda}, \quad a_0 \leq 1. \quad (22)$$

One may tighten $a_0^2 \leq \text{tr}(S)/(\lambda + \text{tr}(S))$. Generate a candidate using $\bar{H}$, and verify each cell with scales $(1 - \eta, 1 + \eta)$ when $\eta < 1$.

Theorem 12 (Response-index sequential repair). *Assume sound response/remainder indices and the declared finite feature contract. Topological candidate generation, shape certification, and eventual exact retained replay return precisely $Q_{\text{seq}}(W, R)$. Updating the intrinsic indices by subtraction and discarding transient calibrated state returns the complete fresh state*

$$\mathcal{A}_{\text{response}}(R) = (Q_{\text{seq}}(W, R), I_{\text{response}}(R), \text{IDs}(R)).$$

The result holds for repeated adaptive requests and permits changed codes at every stage.

Proof. Already certified ancestors determine the true target prefix. Equations (19)–(22) enclose its actual metric. Forced-prefix coordinate induction certifies the emitted tensor; topological induction certifies the model. Complete replay resolves any remaining stage. Additive intrinsic sums and canonical serialization establish state equality independently of request order. $\square$

Quantitative improvement, not just a new notation. In the ideal-exact, exact-jet, in-subspace case $u = 0$, define $K_R^2 = M_0^{-1} \sum_j H_j^2$. Then $\epsilon \leq K_R \|a\|^2 / 2$. If the response candidate's positive-energy normalized margins are at least $\tau > 0$, let $\eta_\tau = \tau / \sqrt{1 + \tau^2}$. Sufficient acceptance is

$$\|a\|^2 < \frac{2\sqrt{\lambda}}{K_R} (\sqrt{1 + \eta_\tau} - 1). \quad (23)$$

Indeed this makes $2e + e^2 < \eta_\tau$, so every displacement is below its margin. If $K_R = 0$, the feature response is exactly affine on the domain and has zero truncation error.

For small τ and controlled constants, the radius is $O(\sqrt{\tau})$; a zeroth-order nonzero-Lipschitz error gives an $O(\tau)$ sufficient radius. This is a local scaling comparison, not universal dominance: margins, conditioning and bound tightness can differ. A first-order residual or nonvanishing finite error floor removes the quadratic improvement. Small deletion fraction alone proves none of these conditions.

20. Adaptive replay, storage, and the true work comparison

For a fixed certified prefix and chart, compute group scalar summaries

$$z_G^2 = \text{tr } S_G(a), \quad \delta_G = 2z_G\epsilon_G + \epsilon_G^2.$$

Then the group's absolute Gram error is at most δ_G . Replay replaces that group's surrogate Gram with the exact target Gram and sets its error to zero. The sum of unreplayed absolute errors decreases. Rewhitening or changing charts need not be monotone. A finite chart bank and eventual replay of a new group ensure termination.

A global root contraction costs $O(r^2d^2)$ arithmetic; ranking G groups from trace/error coefficients costs $O(Gr^2)$. Replacing a group costs an additional contraction unless cached. These counts exclude arbitrary-precision bit costs, parameter projection, provider proofs, index reads, factorization, output and all replay work.

For one chart/stage/group, the minimal coefficient representation stores

$$\binom{r+2}{2} \frac{d(d+1)}{2} \text{ Gram scalars, plus } \frac{(r+3)(r+4)}{2} \text{ error-descriptor scalars.}$$

A small coarse partition and a full leaf tree have very different memory costs. Directions and their domains are additional state. Deleted contributions require regeneration or stored per-record payloads before erasure; IDs alone are insufficient.

Theorem 13 (Coefficient-information lower bound). *Let $K = \binom{r+2}{2}$ and $M = d(d+1)/2$. For every integer $B \geq 2$, some family of integer affine-response corpora needs at least $KM \log_2 B$ bits of stored state to answer every rational surrogate-Gram query exactly, with no record access or other side information.*

Proof. The M matrices $vv^\top$ for $v = e_i$ and $v = e_i + e_j$ span symmetric matrices. The K squares $1, a_i^2, (1 + a_i)^2, (a_i + a_j)^2$ span all scalar quadratics. Their tensor products are independent matrix polynomials. Give each product $B - 1$ fixed record slots; put its affine jet in the first $n \in \{0, \dots, B - 1\}$ slots and zero jets elsewhere. This realizes B^{KM} distinct Grams. Distinct polynomials differ at a rational query, so exact query states must distinguish every member. $\square$

This is a finite-bit bound for polynomial Gram queries, not a universal transformer/model-output lower bound. Degree- p feature surrogates need at most $\binom{r+2p}{2p}$ matrix coefficients; the same construction gives the corresponding bit obstruction, since squares of degree- p polynomials span degree- $2p$ polynomials.

Crucial comparator correction. This index can accelerate an indexed fresh quantizer too. A deletion-exclusive advantage over the best equally indexed algorithm does not follow. The earlier cost ratio applies only when its actual comparator costs F, G, J and all extra repair work are established. Require both ordinary fresh and equally indexed fresh baselines; account for index construction and lifetime amortization. No fair theorem forces the comparator to scan records it can avoid.

21. Lower storage without losing second-order control

The exact quadratic Gram need not be reconstructed to certify a quantization decision. A cheaper tier keeps its constant/linear matrix response and bounds the omitted quadratic matrix using scalar moments.

Put $\Delta Z_j = \sum_t a_t Z_{jt}$, and define normalized retained sums

$$S_{\text{lin}}(a) = M_0^{-1} \sum_j [Z_{j0} Z_{j0}^\top + Z_{j0} \Delta Z_j^\top + \Delta Z_j Z_{j0}^\top],$$

$$Q(a) = M_0^{-1} \sum_j \Delta Z_j \Delta Z_j^\top, \quad G_{st} = M_0^{-1} \sum_j \langle Z_{js}, Z_{jt} \rangle_F, \quad s, t \geq 1.$$

Then $S(a) = S_{\text{lin}}(a) + Q(a)$ and $0 \preceq Q \preceq \beta I$, where $\beta = a^\top G a$. Store $r + 1$ symmetric matrices for S_{lin} and $r(r + 1)/2$ tangent scalars. Also retain the same scalar error-descriptor moments. Matrix storage is $O(rd^2 + r^2)$.

The surrogate feature norm is still exactly accessible: $z^2 = \text{tr } S_{\text{lin}} + \beta$. Thus a feature-remainder bound ϵ gives $\delta = 2z\epsilon + \epsilon^2$ and $-\delta I \preceq H_* - \lambda I - S \preceq \delta I$.

Theorem 14 (Shifted linear-response certificate). *Define $\tilde{H} = \lambda I + S_{\text{lin}} + \beta I$. Then*

$$\tilde{H} \succeq \lambda I, \quad -(\beta + \delta)I \preceq H_* - \tilde{H} \preceq \delta I.$$

If $\beta + \delta < \lambda$, valid relative scales are

$$a_- = 1 - \frac{\beta + \delta}{\lambda}, \quad a_+ = 1 + \frac{\delta}{\lambda}. \quad (24)$$

The exact cell verifier with these scales certifies the original target. In the exact in-chart jet regime, both β and δ are $O(\|a\|^2)$, so the second-order uncertainty survives.

Proof. Since $\beta I - Q \succeq 0$, $\tilde{H} = \lambda I + S + (\beta I - Q) \succeq \lambda I$. Subtract this identity from the target and combine the PSD omission bound with the feature remainder. The ridge floor converts absolute to relative bounds. The order follows because ΔZ is linear in a and the feature remainder is quadratic. $\square$

An unshifted $H_{\text{lin}} = \lambda I + S_{\text{lin}}$ has automatic floor $\lambda - \beta$ when positive. It is another valid candidate, not a universally better one. Its own margins must be checked.

Replay preserves a useful invariant. Shift each unresolved group by $\beta_G I$. Every raw group proposal is PSD. After replay replaces any proposal by its exact Gram, the global ridge floor remains λ . With $A = \sum_{\text{unresolved}} (\beta_G + \delta_G)$ and $D = \sum_{\text{unresolved}} \delta_G$, the scales are $(1 - A/\lambda, 1 + D/\lambda)$. These particular conservative endpoints tighten monotonically during replay; all replay recovers the exact target.

An implementation-matched zero-replay condition is available: if every stage's shifted candidate has normalized margin $\tau > 0$ and

$$\beta + \delta \leq \lambda/2, \quad \beta + 2\delta < \lambda\tau,$$

then $\chi \leq (\beta + 2\delta)/\lambda < \tau$, so every stage accepts. This permits changed codes throughout. Finite/residual terms remain in δ ; no real-model coverage follows without their bounds.

The earlier storage lower bound required exact polynomial-Gram queries. This tier answers an enclosure query, so it escapes that bound without contradiction. The gain is reduced stored matrices, not a proved wall-clock or certificate-coverage improvement.

22. Transformer curvature and executable coverage

For an ideal graph node $x(a)$ maintain region-wide bounds (M_x, J_x, H_x) on value, full coefficient Jacobian and full bilinear coefficient Hessian. These are proof bounds over a region, not derivatives sampled at its center.

For affine $W(a)$ and $y = W(a)x(a) + b(a)$,

$$J_y \leq M_W J_x + J_W M_x + J_b, \quad H_y \leq M_W H_x + 2J_W J_x.$$

For a matrix product $y = xz^\top$,

$$H_y \leq H_x M_z + 2J_x J_z + M_x H_z.$$

For $y = g(x)$ with derivative bounds L, B ,

$$J_y \leq L J_x, \quad H_y \leq L H_x + B J_x^2.$$

The mixed terms are essential when multiple ancestors change. For ideal row softmax, valid constants are $L = 1/2$, $B = 3/2$: differentiate $J = \text{diag}(p) - pp^\top$ to obtain $DJ[k] = \text{diag}(Jk) - (Jk)p^\top - p(Jk)^\top$ and use $\|p\|_2 \leq 1$. Frobenius row aggregation preserves the constants.

For LayerNorm of width n , fixed gain norm Γ and region denominator lower bound $s_{\min} > 0$, valid constants are

$$L \leq \Gamma/s_{\min}, \quad B \leq 6\Gamma/(\sqrt{n} s_{\min}^2).$$

Exact GELU admits conservative global constants $L < 1.25$, $B \leq 2$. Residual merges add bounds. The independent derivation supplies full formulas. Finite kernels require the separate numerical provider; these ideal inequalities do not certify arbitrary libm/CUDA operations.

Executable component	Scope and present limitation
Exact quantizer	Fraction reverse LDL and fixed-grid decisions; independently checked against constrained Gaussian solves.
Sparse injection	Target factor known; interval premises explicitly labeled; exact ambiguous-coordinate fallback.
Response moments	Exact quadratic/linear contractions and retained-only subtraction; supplied jet/remainder provenance remains a premise.
Stage service	Complete fresh/repair paths, bound-to-prefix witnesses, selected replay and canonical commit. Trusted evaluator/provider boundary is explicit.
Decoder backend	Complete scalar CPU decoder with causal attention, LayerNorm, GELU, residuals and logits. Fixed architecture and supplied weights/token IDs.
Numerical shortcut	Legacy decoder certifies structural identity. Revision 4 adds automatic interval jets and finite-error proofs for a separate certified decoder.
Work scheduler	Cooperative bounded packets, cancellation and commit counted; not a GPU preemptor or latency measurement.

23. Contribution discipline and publication gates

Recommended central claim. Exact calibration deletion for a sequential quantizer through deletable response moments, certified discrete decisions, repeated canonical state and explicit replay/storage costs. The response index’s second-order local regime makes this stronger than an unchanged-model gate. Its practical usefulness is still to be measured.

The new audit found material adjacent prior work. We therefore avoid broad priority claims:

[7] Lin and Sinnamon. *The Generalized Wielandt Inequality in Inner Product Spaces*. 2012.

<https://arxiv.org/abs/1201.6294>. Our shape constant is a specialization of classical angle/conditioning geometry.

[8] Gunn. *How to sketch a learning algorithm*. April–June 2026.

<https://arxiv.org/abs/2604.07328>. This gives derivative-based sketches for approximate deletion-output prediction under stability. Derivative sketches for deletion are already established; our proposed distinction is exact sequential discrete output and canonical intrinsic state.

[9] Shen/Li. *Published Unlearning Numbers Move Per Checkpoint, and Not Because the Removed Data Survives*. September 2026. <https://arxiv.org/abs/2609.11490>. Appendix E audits retained-only calibration-scale refitting at fixed weights in vision models. Awareness of calibration influence is not novel by itself.

[10] Entesari and Fazlyab. *Hierarchical End-to-End Taylor Bounds for Complete Neural Network Verification*. L4DC 2026. <https://proceedings.mlr.press/v331/entesari26a.html>. Certified neural Taylor bounds are prior machinery.

[11] Huggins et al. *PASS-GLM: Polynomial Approximate Sufficient Statistics for Scalable Bayesian GLM Inference*. NeurIPS 2017. <https://arxiv.org/abs/1709.09216>. Polynomial sufficient statistics are established; the sequential quantization certificate and deletion state are the proposed application-specific contribution.

The full dated audit and additional primary sources are in `docs/NOVELTY_AUDIT.md`. This is a focused audit, not proof that no matching work exists.

What is closed. The stated exactness, remainder, canonical-state, coefficient-information and conditional charged-work results have derivations and independent review. A complete reference execution path now exists, including a deterministic decoder and exact fallback. Software correctness tests are recorded separately from experiments.

What remains open. Useful chart coverage, competitive storage/preparation, and reliable real-model latency/NLP evidence remain unmeasured. Strict gain over equally indexed fresh evaluation is not automatic. UNKNOWN-to-replay ensures safe fallback, not speed. Revision 4 supplies compact aggregate state and a certified finite provider. It also loads local GPT-2 safetensors. These are scalar reference implementations. They are not production GPU kernels or evidence of practical speed.

Next experiment gate, when authorized. Validate full artifact and repeated-state equality independently, then measure complete service cost, real text quality, retained replay, storage and amortization. Include failed certificates, changed first layers, an equally indexed fresh baseline and a fixed-reference ablation. Continue deferring synthetic empirical datasets. No theory-only promise replaces those measurements.

24. Compact complete state is now executable

The transparent response adapter keeps per-record matrices. Revision 4 adds a separate complete service with group sums. Its committed state has three parts:

$$\mathcal{A}_{\text{agg}}(R) = (Q_{\text{seq}}(W, R), \text{Bindings}(R), \{\text{LinearMoments}_g(R), \text{ErrorMoments}_g(R)\}_g).$$

Each record binding contains its identifier, payload digest, fixed group, and stage-contribution digests. It contains no feature matrix or derivative array. Each group stores exact response sums, scalar remainder moments, and unavailable-evidence counts. Embedded aggregate indices have empty per-record payload lists.

Deletion update. Recompute each deleted record’s intrinsic moments from its supplied payload. Verify its binding before subtraction. Subtract those exact contributions from group totals. Missing proof evidence uses a canonical marker. Any unresolved group then uses exact retained replay.

Proposal query. Read the aggregate moments and the installed certified prefix. Solve the fixed chart-membership problem. Contract the group response and error moments. Construct the shifted PSD proposal from Section 21. Bind every enclosure to the actual prefix and proposal. The existing decision verifier either certifies all decisions or requests further replay.

Theorem 15 (Compact-state equivalence). *Assume intrinsic extraction is deterministic and independent of the deletable corpus. Assume every claimed enclosure is sound and required finite evaluations succeed. Then compact repair returns the fresh retained model and canonical compact state. This holds for repeated deletions in any order.*

Proof. Exact additive subtraction leaves precisely the retained intrinsic sums. Fixed group assignment and sorted bindings give the same serialization as fresh construction. Unavailable counts also subtract exactly. At each stage, accepted cells contain the true retained target. Replay instead supplies its exact Gram. Induction therefore gives the fresh stage outputs. The final state depends only on retained records and fixed configuration. $\square$

Honest storage and work. Matrix storage is independent of record count for fixed groups, chart rank, and model dimensions. The response tier uses $O(G(rd^2 + r^2))$ rational entries per stage. Scalar remainder moments add $O(Gr^2)$ entries. Membership and contribution digests still use $O(NL)$ metadata. Validation, membership rebuilding, and canonical serialization can scan this metadata. Deleted-record extraction and all replay reads remain charged. Arbitrary-precision integer sizes also matter.

The service retains no request-local record, feature, or derivative cache after return. This is a logical committed-state guarantee. It does not promise physical erasure of Python memory or authentication against hostile storage.

25. A concrete finite provider for changed ancestors

Revision 4 defines V_{cert} as a separate numerical target. Basic operations use declared binary64 round-to-nearest-even semantics with gradual underflow. Nonlinear primitives use rational enclosures. Refinement continues until both endpoints select the same binary64 result. An unresolved rounding decision aborts the finite execution. It never returns an uncertified approximation.

Square roots use integer-square inequalities. Exponentials use range reduction and an explicitly bounded Taylor tail. Error-function bounds use series and tail inequalities. Hyperbolic tangent uses certified exponential identities. Only data-independent constants may be cached. The source and runtime manifests distinguish this program from the earlier library-math decoder and floating quantizer E .

Automatic proof construction. Fix affine ancestor-weight directions independently of calibration records. Let a lie in a fixed convex box. Interval second-order differentiation bounds values, every first derivative, and every mixed second derivative. A separate error component bounds finite execution against the ideal smooth graph. The chart is checked against the installed binary64 weights. Any unexplained residual returns UNKNOWN.

For one finite basic operation, a sufficient rounding bound is

$$|\text{fl}(z) - z| \leq 2^{-53}|z| + 2^{-1075},$$

provided overflow is excluded. Arithmetic propagation adds existing operand errors before applying this bound. Nonlinear propagation uses derivative bounds over expanded input intervals. Denominators must remain separated from zero. Possible overflow, invalid domains, or unresolved interval calculations reject the proof.

Stable softmax. The finite evaluator subtracts the maximum score before exponentiation. The proof does not differentiate that discrete branch. It bounds ideal softmax derivatives directly and separately encloses the finite max-shift schedule. This distinction preserves smooth Hessian bounds and accounts for actual rounding.

For record j , choose interval-midpoint feature jets at the chart center. Let e_{j0} bound their constant error. Let e_{jt} bound derivative errors. Let H_j bound the full Hessian tensor. Let ν_j bound finite execution error. Then

$$\|X_j^{\text{finite}}(a) - Z_{j0} - \sum_t a_t Z_{jt}\|_F \leq \nu_j + e_{j0} + \sum_t |a_t| e_{jt} + \frac{1}{2} H_j \|a\|_2^2.$$

The stored squared scalar descriptors include every cross term. Their aggregate contraction feeds the signed Gram enclosure from Section 21. No user-supplied numerical tolerance replaces this derivation. Proof failure causes replay. Failure inside the finite evaluator aborts the request without an approximate result.

This implementation closes the automatic-provider interface for its declared program and accepted chart domain. It does not establish wide coverage. Interval dependency, large charts, and finite-error accumulation can cause abstention. Full-dimensional charts can also make preparation and storage prohibitive.

26. Checkpoint interface and remaining boundaries

The local adapter reads GPT-2 configuration and safetensors without downloads or pickle. It supports single files and shards. It decodes F16, BF16, F32, and F64 weights. Conv1D matrices are transposed into the reference decoder’s convention. Configuration checks cover attention scaling, normalization, activation, and output-head handling.

The adapter supports the declared sequential pre-normalization GPT-2 architecture. It rejects cross-attention, pruned heads, unsupported attention variants, and inconsistent tensors. A bare GPT2Model import explicitly declares its added output head. Source-file hashes and mapping details enter provenance. Imported weights do not establish native Hugging Face numerical equality.

Wrap the imported decoder with the certified evaluator to select V_{cert} . Supply corpus-independent chart directions, fixed grids, ridge, and normalization. These choices define the target before calibration. Unsupported chart changes replay retained data. A checkpoint-derived chart may use base weights. A chart fitted to deletable calibration data needs its own deletion semantics.

Implementation question	Revision 4 answer
Compact complete service	Group sums replace persistent per-record matrix payloads. Meta-data scans remain explicit.
Automatic changed-prefix bounds	Interval jets and finite-error propagation supply concrete descriptors for the certified decoder.
Stable finite primitives	Rational refinement supplies certified rounding or an explicit failure.
Local checkpoint loading	GPT-2 safetensors mapping is implemented and checked with software fixtures.
Native checkpoint equivalence	Not claimed. The certified numerical program has its own manifest.
Practical large-model execution	Not established. Eager scalar loading and interval Hessians can be costly.
Real-data speed and quality	Unmeasured. Research experiments remain paused.

Validation boundary. All 129 correctness tests pass. Independent review found no unresolved defect under the declared contracts. One fixture deletes a record and changes an early QKV code from zero to $1/1024$. Its downstream features also change. Certified repair reads no retained records and matches fresh canonical state. Downstream matrices mostly start on-grid. This fixture does not establish practical coverage. The test inventory and source hashes appear in `docs/VALIDATION.md`.

Next authorized phase. First audit exact outputs and all fallback paths on local real checkpoints. Then measure complete service cost and certificate coverage. Charge chart preparation, deleted-record extraction, metadata scans, retained reads, output writes, and storage. Compare against an equally indexed fresh solver. Keep failed requests and changed early weights in the workload. No universal or reliable measured speedup follows from this implementation alone.